



**UTM**  
UNIVERSITI TEKNOLOGI MALAYSIA

# **SECV2113 HUMAN COMPUTER INTERACTION**

**SEMESTER 2 (2023/2024)**

**SECTION 02**

## **ASSIGNMENT 2**

**HEURISTIC EVALUATION**

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**DATE OF SUBMISSION:** 27<sup>th</sup> JUNE 2024

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## Introduction

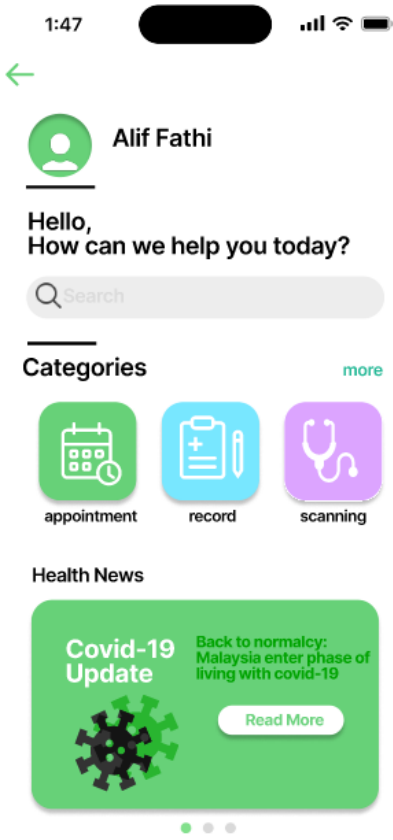
Heuristic Evaluation (HE) is a systematic approach for assessing the user interface (UI) of an application, guided by a set of established principles or heuristics. In the context of our second assignment, we are tasked with conducting an evaluation of an application named HealthMax, developed by the #KAMIXRAMAI group. Our analysis of the app's usability will be grounded in Nielsen's heuristics, a comprehensive set of ten principles designed to facilitate effective user interaction. These principles serve as our roadmap for the evaluation process. Here are the 10 Nielsen's heuristics:

1. Visibility of system status. - keep users informed about what is going on in real-time
2. Match between system and real world. - use language, with words, phrases and concepts from real life that is familiar to the user
3. User control and freedom. - users often choose system functions by mistake and will need a clearly marked "emergency exit"
4. Consistency and standards. - users should not have to wonder whether different words, situations, or actions mean the same thing.
5. Error prevention. - design that prevents a problem or error from occurring
6. Recognition rather than recall. - make it simpler for users to recognize and access the system
7. Flexibility and efficiency of use. - provide immediate and effective solutions for better user experience
8. Aesthetic and minimalist design. - should not contain information which is irrelevant or rarely needed
9. Help users recognize, diagnose, and recover from errors. - error messages should be expressed clearly and suggest a solution
10. Help and documentation. - any such information should be easy to search

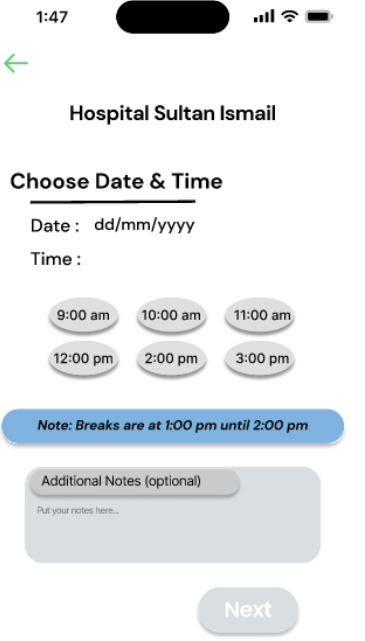
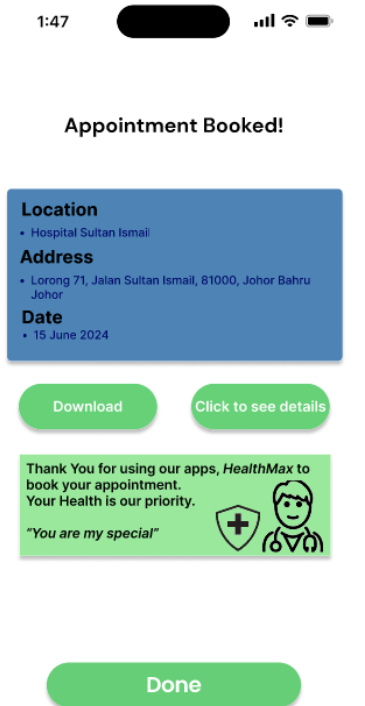
Each heuristic can be categorized into different severity levels (1-4):

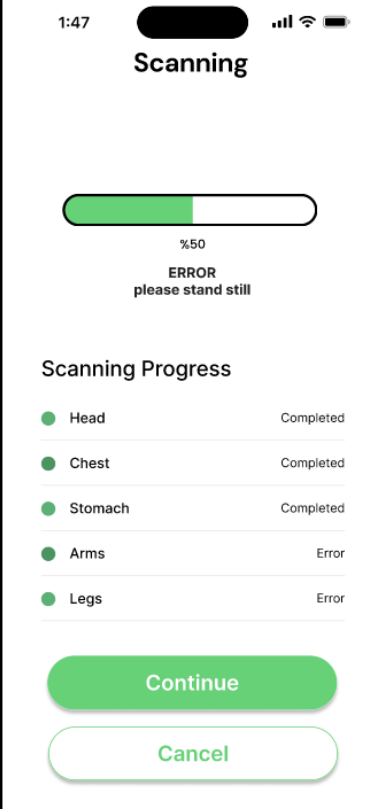
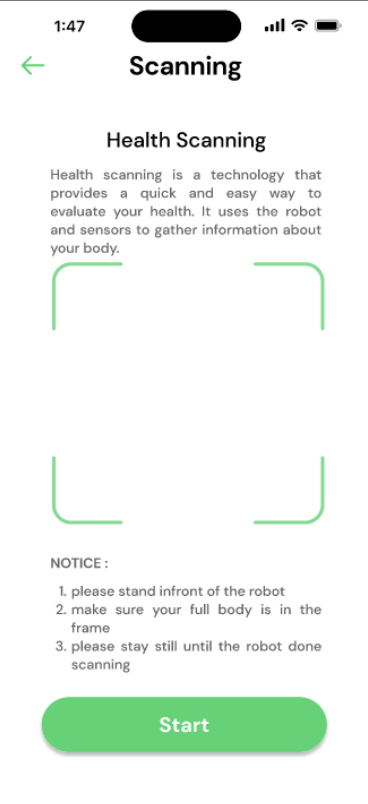
1. Cosmetic issue – affects the appearance and should be fixed only if time permits
2. Minor issue – hinders the user's ability to navigate and should be fixed when
3. Major issue – Frustrates or confuses users and requires repair as soon as possible
4. Catastrophic issue – prohibits users from performing their given task and requires an immediate modification

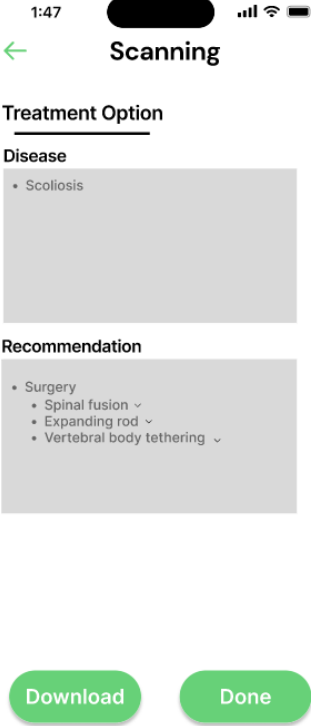
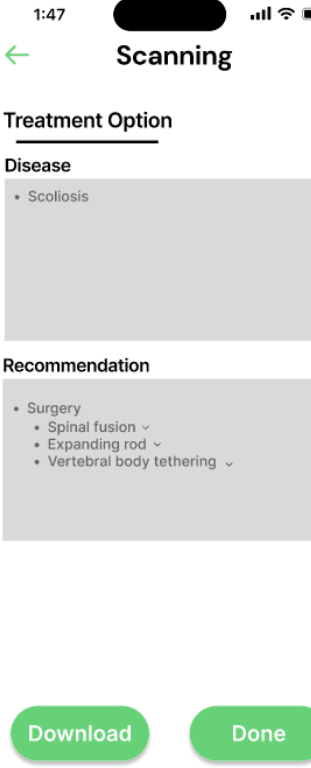
## Heuristic Evaluation

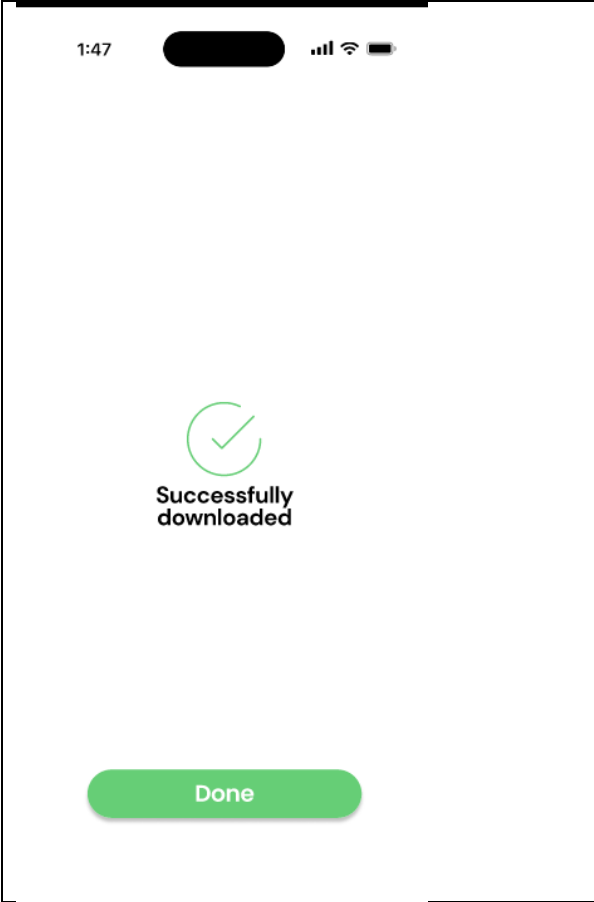
| Prototype Image                                                                    | Identified Issue                                                                                                                                                                                                                                                                                                                                 | Heuristic and Severity                                                |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
|  | <p>The labels used for the features in the app may not be as intuitive to some users. Instead of "appointment", "record", and "scanning", the labels could be "Book appointment", "Medical Record" and "Health Scan".</p> <p>The "Categories" can also be changed to "Services" to show that the icons indicate a usable feature on the app.</p> | <p>H2: Match between system and real world</p> <p>S2: Minor issue</p> |

|                                                                                    |                                                                                                                                                                                                                                                                    |                                                             |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
|  | <p>The “Click here” buttons underneath both Non-Emergency and Immediate Treatment can be more descriptive and specific. They can use a “Book Appointment” or “Book Now” to provide clearer information about what will happen when the user clicks the button.</p> | <p>H4: Consistency and Standards</p> <p>S2: Minor Issue</p> |
|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                              |                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
|  <p>1:47</p> <p>←</p> <p><b>Hospital Sultan Ismail</b></p> <p><b>Choose Date &amp; Time</b></p> <p>Date : dd/mm/yyyy</p> <p>Time :</p> <p>9:00 am 10:00 am 11:00 am<br/>12:00 pm 2:00 pm 3:00 pm</p> <p>Note: Breaks are at 1:00 pm until 2:00 pm</p> <p>Additional Notes (optional)<br/>Put your notes here...</p> <p>Next</p>                                                                                                | <p>There is no indication as to which time slot is available. This can cause error and confusion and possibly the appointment overlapping with someone else's appointment. The unavailable time slots can be grayed out.</p> | <p>H1: Visibility of System Status</p> <p>S4: Catastrophic issue</p>                              |
|  <p>1:47</p> <p><b>Appointment Booked!</b></p> <p><b>Location</b><br/>• Hospital Sultan Ismail</p> <p><b>Address</b><br/>• Lorong 71, Jalan Sultan Ismail, 81000, Johor Bahru<br/>Johor</p> <p><b>Date</b><br/>• 15 June 2024</p> <p>Download Click to see details</p> <p>Thank You for using our apps, <i>HealthMax</i> to book your appointment.<br/>Your Health is our priority.<br/>"You are my special"</p> <p>Done</p> | <p>It is unclear as to whether it is possible to cancel the appointment or modify certain details of the appointment.</p>                                                                                                    | <p>H1: Visibility of system status</p> <p>H3: User Control and Freedom</p> <p>S3: Major issue</p> |

|                                                                                     |  |                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                               |
|-------------------------------------------------------------------------------------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
|   |  | <p>It is hard to identify what is causing the error on first glance. The “ERROR please stand still” message can be more specific, like “Please keep your arms on the side of your body and straighten your legs” The Arms and Legs error can be made to be more prominent to the user by using red for the “error” font or highlighting it with red.</p> | <p>H1: Visibility of System Status<br/>H9: Help users recognize, diagnose, and recover from errors</p> <p>S3: Major Issue</p> |
|  |  | <p>The scan box looks like a QR scanner even though it is stated that you must stand in front of the robot to get a full body scan below the scan box. This may get some users confused, and cause misunderstanding.</p>                                                                                                                                 | <p>H4: Consistency and Standards</p> <p>S3: Major issue</p>                                                                   |

|                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                       |
|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
|   | <p>Some users may not be familiar with medical terms like the disease that they have or the recommended treatment that they have to take. It is good to provide a description to let the users know</p>                                                                                                                                                                                                        | <p>H2: Match between system and real world</p> <p>S2: Minor issue</p> |
|  | <p>It is unclear as to what “Download” and “Done” button will do. Users may think that clicking the “Done” button will save their scanning history with the option to download the report, or they may think that it will be automatically downloaded locally. On the other hand, the users may think that clicking the “Download” button will automatically end their session after the download is done.</p> | <p>H4: Consistency and standards</p> <p>S1: Cosmetic issue</p>        |

|                                                                                    |                                                                                                                                                                                                                                                                                                                                           |                                                               |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
|  | <p>It is unclear as to what was downloaded, and what is the content of the report and where it is located after the download process is done. The app can instead make a “View Report” button, and a short description under the “Successfully downloaded” message saying “Your Health Scan report has been successfully downloaded.”</p> | <p>H1: Visibility of system status</p> <p>S3: Major issue</p> |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|

User Journey Map

MUHAMMAD SUMBUL

Muhammad Sumbul is having a fever today but he has to go to work. He wants to check the severity of his condition and what disease does he have and see if he has to do an immediate check up or stay at home while doing work.

EXPECTATIONS:

- Easy and smooth interaction with the health app
- Clear informaton: disease, treatments, temperature, heart rate, blood pressure
- Ability to provide accurate information of health scanning and treatment options

